Cohort 1 - the clinical catalogue, measured

What a clinical-only selection rule actually produces against ChEMBL 37, and how well evidenced the molecule-target claims inside it are.

Revised 2026-09-12 after review of 11 Zotero questions. Definitions are expanded in place; exclusion retention and policy counts are corrected. The original annotated edition is preserved in Zotero, with an unchanged PDF, Markdown, and annotation export in `docs/archive/cohort-report-2026-09-12/`. The headline below remains the original ChEMBL 37 baseline. New recovery and policy comparisons are reported separately in section 6 and the follow-up analysis; they do not silently replace it.

This report chooses nothing. It measures the consequences of the clinical option in docs/MOLECULE-EXPANSION-BRAINSTORM.md so a selection policy can be chosen with numbers in hand. Nothing here changes the seed list, the schema, the database, the graph export or the viewer, and the tiers below are **report definitions, not production trust rules** - the invariants in the project guide are untouched.

Cohorts 2 (curated ligands) and 3 (broad experimental) are deferred and not compared here.

The original scripts and saved outputs are in `analysis/cohort1/`. The original MeSH cache files are absent from this checkout, so `run_all.sh` would fetch a new MeSH snapshot rather than reproduce its bytes. The follow-up reruns offline from the saved descriptor classification and the pinned ChEMBL release: `python3 analysis/cohort1/followup/analyze.py`.

The headline

A clinical-only rule - ATC N **or** an in-scope CNS indication, deduplicated to chemical parents - admits **2,245 molecules** from ChEMBL 37, of which **1,771 are small molecules**. That is **80× the current 28-molecule seed set**, and it reaches **4,885 distinct targets** (3,750 UniProt accessions) against the graph's present 207.

Here **chemical-parent deduplication** counts related salt/hydrate records once while retaining their original identities; see the worked theophylline example in section 2. It does not merge distinct prodrugs or stereoisomers where ChEMBL gives them separate parents.

The evidence behind it is thinner than the size suggests, and thins further the more carefully the target assignment is read:

	Claims	Molecules
Molecule-target claims assessed	138,509	2,245
Supported	9,014 (6.5%)	1,555
Preliminary	38,506 (27.8%)	-
Limited or unresolved	90,989 (65.7%)	-
Supported on **direct human single-protein** evidence only	3,442 (2.5%)	1,125
No eligible evidence under the report rules (ChEMBL 37)	-	361

Preliminary means experimental evidence is available for that particular molecule-target-role claim but does not meet the report's replication rule. It is not a clinical phase or a rating of the whole molecule; section 3 gives the study and target-assignment qualifications.

Three findings drive every recommendation at the end:

1. **Two thirds of all claims are Limited or unresolved. 59,327 claims (43% of the total) have no qualifying exact value and at least one bound or approximate result.** These are informative observations, but cannot be pooled as exact measurements. Their direction matters: `Ki > 10,000 nM` and `Ki < 1 nM` imply different things, and ~ means approximate rather than censored. Preserving that distinction extends invariant 1 to a much larger dataset.
2. **The two Supported routes barely overlap.** 1,321 molecule-target pairs are supported by curation alone, 6,138 by replicated measurement alone, and only **538 by both**. Curation and measurement are answering different questions, and a single "supported" badge would hide that.
3. **The catalogue is defined independently of the graph and it shows.** 206 of the current 207 targets reappear, and 3,544 accessions are new. The clinical rule does not merely extend the caffeine neighbourhood; it replaces its centre of gravity with the aminergic receptors and transporters.

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1. How membership is defined

Two admission routes, kept separate throughout. **Neither is evidence of anything.** ATC N is a classification, not a verified claim of central action; an indication record is a development or approval annotation, not a target relationship. Membership is settled first, and evidence is assessed afterwards on the fixed candidate set - so evidence availability never decides who is in the catalogue.

Route A - ATC N

A molecule is admitted when it, or any member of its ChEMBL compound family, carries an ATC level-5 code whose level 1 is `N` (nervous system).

ATC is a hierarchy of drug categories. A **subtree** means a branch plus its more specific categories: for example `N` (nervous system) contains `N01` (anaesthetics), which contains `N01B` (local anaesthetics).

ATC N is taken **whole** in this baseline. The report-generation instructions asked for explicit decisions about local anaesthesia and general pain, among other boundaries. The analysis author translated those concerns into two ATC branch flags: they are *flagged in the output rather than removed*. This was a report design choice, not a requirement to exclude them and not a production selection rule.

Flag	ATC level 3	Molecules	Why flagged
`peripheral_site_of_action`	N01B - anaesthetics, local	20	Local anaesthetics block peripheral nerve conduction at the site of application. ATC files them under N because they act on nervous tissue, not because they act centrally.
`general_analgesic`	N02B - other analgesics and antipyretics	37	Paracetamol and kin. Same argument as the pain indication class, arriving by the ATC route.

They are flagged, not dropped, precisely because the ATC route's whole point is that it is a classification and not a verified claim - hiding its weakest cases would make it look stronger than it is. Several (lidocaine, mexiletine) have real central and cardiac target pharmacology anyway.

Route B - CNS indication scope

A molecule is admitted when a `drug_indication` record for it, or for a family member, names a MeSH descriptor inside the report's saved indication scope.

MeSH means *Medical Subject Headings*, the US National Library of Medicine's controlled vocabulary for organising biomedical information. A **descriptor** is a defined concept with an identifier, such as `D010146` for Pain. A **versioned scope** is our documented rule for which concepts qualify: this report uses version **1.0.0** in `config/cns_indication_scope.json`. MeSH supplies the vocabulary; the project chooses the inclusion rule. The rule version and the dated MeSH snapshot are separate parts of reproducibility. NLM MeSH

MeSH tree numbers are addresses within that hierarchy. `C10` identifies the nervous-system disease branch; `C10.597.617.470` is one placement of Neuralgia. One descriptor may have several placements. Its descriptor ID identifies the concept, while its tree numbers locate it within categories. NLM tree structures

The analysis obtains those placements by exact descriptor ID from the NLM MeSH SPARQL endpoint, a service for structured queries. Heading text is not the final matching rule. A name search can return a related concept, synonym, or broader term; it does not always fail. The project's actual ADRA1A-to-ADRA1D search error occurred in UniProt, not MeSH, and motivates invariant 15.

All **2,178** distinct `mesh_id` values in ChEMBL 37's `drug_indication` resolved in the original run. **Unresolvable** would mean an ID could not be matched to its MeSH record and tree information. Zero failed mappings checks that lookup, not the biological validity of the chosen scope.

Descendant rule. A descriptor is in a class when any of its tree numbers equals the class tree number or begins with it followed by a period.

Core scope = `C10` (Nervous System Diseases) union `F03` (Mental Disorders), minus the excluded boundary classes below. 974 descriptors, 359 of them actually used by ChEMBL indications.

Exclusion rule. A descriptor leaves the core only when *every* one of its in-scope tree placements sits inside an excluded class. Anything with a remaining in-core placement stays. This is what keeps myasthenia gravis (autoimmune `C10.114` **and** neuromuscular-junction `C10.668.758`) inside the core while genuinely muscle-only disease falls out.

Boundary decisions, with the argument against each

Class	MeSH	In core?	Reason	Counter-argument
CNS oncology	`C10.551`, `C04.588.614`	**No**	Sits inside C10 but admits drugs selected by oncology pharmacology. The relationship explaining a temozolomide glioma record is DNA alkylation, not neurotransmission.	Agents used for tumour-associated seizures or oedema are genuinely central - and they re-enter the core through their own non-oncology indications.
Peripheral / cranial nerve	`C10.668.829`, `C10.292`	**No**	A peripheral neuropathy indication does not establish central action, which is the property the catalogue is about.	Pregabalin, duloxetine and carbamazepine act centrally - and nearly all re-enter through epilepsy, mood or neuropathic-pain descriptors carrying a `C10.597` placement.

Class	MeSH	In core?	Reason	Counter-argument
Muscle and NMJ	`C10.668.491`, `C10.668.758`	**No**	Muscular dystrophies are diseases of muscle, filed under C10 only because MeSH groups neuromuscular disease there.	Cholinesterase inhibitors for myasthenia gravis share targets with the Alzheimer's drugs central to this project - and they are admitted through the autoimmune placement anyway.
General pain	`C23.888.592.612`	**No - by construction**	*Measured, not assumed*: MeSH `Pain` (D010146) carries `C23.888.592.612`, `F02.830.816.444` and `G11.561.790.444`, and **none under C10 or F03**. General pain never enters the core; no rule is needed to keep it out.	`Neuralgia` (D009437) carries **both** `C10.597.617.470` and `C23.888.592.612.547`, so nervous-system pain is already in core while non-neurological pain is not. This is the split worth keeping.
Local anaesthesia	-	n/a	Not a disease; it reaches the catalogue only through ATC N01B, and is flagged there.	-

Each class is reported as its own additive delta, so none of these is a final decision. Full per-descriptor detail: `out/indication_scope.csv`.

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2. What the routes produce

Admission-route arithmetic

Deduplicated to chemical parents. `arithmetic_holds` is verified for every row (`|A union B| = |A| + |B| - |A intersection B|`) - check 3 in `out/validation.csv`.

Stage	ATC N only	Indication only	Both	Union
All molecule types	136	1,701	408	**2,245**
Small molecules only	132	1,250	389	**1,771**
Small molecules with an Approved record in the family	65	591	296	**952**

The routes are **not** substitutes. Only 408 molecules arrive by both; 1,701 arrive by indication alone and would be invisible to an ATC-only rule, and 136 arrive by ATC alone with no in-scope indication record at all.

Checking the earlier ATC-N counts (ChEMBL 37)

The **baseline** is the earlier ATC-N-only calculation in the brainstorming document. **Reproduction** means rerunning that same query, with the same parent normalisation, small-molecule filter and ChEMBL release, and checking that the counts agree. It does not mean independently repeating experiments or validating every conclusion in this report. The counts agree **exactly**:

Quantity	Document	Measured
ATC N small-molecule parents	521	**521**
...with an approved record in the compound family	361	**361**
Current seed molecules inside ATC N	15 of 28	**15 of 28**

No discrepancy to explain. The differences that appear elsewhere in this report come from deliberately different scope - 544 rather than 521 because the catalogue does not pre-filter to small molecules, and 373 rather than 361 for the same reason.

Scope sensitivity

Variant	Union	Δ vs core
S0 core (C10 + F03 - excluded classes)	**2,245**	-
S1 + CNS oncology	2,323	+78
S2 + peripheral / cranial nerve	2,330	+85
S3 + muscle and NMJ	2,338	+93
S4 + general pain	2,445	+200
S5 widest (all of C10 + F03 + Pain)	2,611	+366

The boundary decisions move the catalogue by at most 16%. General pain is the largest single one at +200 - and it is also the one whose molecules are least likely to be explained by a central target relationship.

Clinical stage

Two different quantities, reported in two columns and never merged.

Indication-specific (`drug_indication.max_phase_for_ind` for the qualifying CNS indication) - the honest one for an indication-based selection:

Stage for the qualifying CNS indication	Molecules	Cumulative
Approved (ChEMBL 4)	506	506
Phase 3	491	997
Phase 2	769	1,766
Phase 1	311	2,077
Early Phase 1 (0.5)	25	2,102
Unknown (ChEMBL -1)	7	-
Not applicable (ATC route only, no in-scope indication)	136	-

Molecule-level (`molecule_dictionary.max_phase` across the compound family, any condition): 1,147 Approved, 386 Phase 3, 591 Phase 2, 101 Phase 1, 19 Unknown.

The gap between them is the point. **1,147 candidates have an Approved record somewhere in the family, but only 506 are approved *for a CNS indication*.** 591 are approved elsewhere while their qualifying CNS indication sits below phase 4 - prazosin (`CHEMBL2`) is approved, and Phase 3 for its CNS indication. Collapsing those columns would have inflated "approved CNS drugs" by more than twofold.

ChEMBL's `max_phase = 4` labels **Approved**. It is not proof of current approval in any particular jurisdiction, and it is not a completed Phase IV trial. Missing and Unknown (-1) are kept as distinct categories throughout, never folded into zero. Withdrawal annotations are preserved: **78** candidates carry `withdrawn_flag` somewhere in the compound family.

Molecule categories

Small molecules are the main comparison; everything else is accounted for separately rather than filtered away.

molecule_type	**Molecules**	structure_type	**Molecules**
Small molecule	1,771	MOL (a structure)	1,761
Protein	127	SEQ (sequence only)	232
Unknown	117	BOTH	30
Antibody	115	**NONE (unresolved identity)**	**222**
Enzyme	32		
Oligonucleotide	31		
Gene	19		
Vaccine component	16		
Oligosaccharide	11		
Cell	5		
Antibody drug conjugate	1		

222 candidates have no chemical structure in ChEMBL at all (`structure_type = NONE`) - mixtures, botanicals such as `GINKGO`, and formulation records. They are retained with their identity marked unresolved. No endogenous ligand, research tool or active metabolite was added; every row here met a clinical selection rule on its own.

Identity: parent/salt normalisation

Deduplicated to chemical parents means counting related salt or hydrate forms once, using ChEMBL's explicit parent links. A parent is the reference compound in this database hierarchy, not a biological precursor. The analysis uses the parent's `molregno` identifier while every contributing source form is kept in `out/family_members.csv`. Nothing is matched on a name.

This prevents several forms of one compound appearing as several catalogue molecules. It does not assert identical absorption or exposure across formulations, and mass-to-molar conversions must still use the chemical form actually measured rather than automatically using the parent freebase weight.

- 2,245 parents span 3,668 family members - **1,423 salt, hydrate and formulation records folded in**.

- 883 parents have more than one form; the largest family has 16.
- **11** family members carry the ATC N code while the parent does not, and **448** carry the qualifying indication while the parent does not. Ignoring the hierarchy would have lost those molecules entirely.
- Meaningful distinctions are preserved: stereoisomers, prodrugs, metabolites and mixtures stay separate parents where ChEMBL treats them so.

The known hard case behaves correctly. The seeded theophylline record `CHEMBL1355736` (the monohydrate - and typed `Protein` in ChEMBL 37, a source oddity worth noting) rolls up to parent `CHEMBL190`, "theophylline anhydrous". It is **one** catalogue row, not two. This is the same record the data-quality log records as a false positive in an earlier sweep, and it is the only seed that is not its own parent.

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3. Target evidence

What was assessed, and what it excludes

Uniformly across all 2,245 candidates and their 3,668 family members: every `drug_mechanism` record and every `activities` row in ChEMBL 37.

1,138,364 activity rows scanned. 310,719 fell in scope (assay type B or F, an endpoint type the report recognises); **117,519 were valid exact measurements**.

Excluded / retained	Activity rows	In baseline measurement audit?
Assay type not B or F	483,915	no
Standard type out of scope	321,727	no
Bound or approximate result (>, <, ~, ...)	**81,867**	**yes, relation preserved**
Null value	102,708	yes, excluded from numerical use
Organism-based format (whole-organism readout)	21,999	no
Other relation	5,903	yes, excluded from numerical use
Units not nM	2,714	yes, original value and units
Non-positive value	8	yes, excluded from numerical use
Documented known-bad records	**4**	**no; aggregate exclusion count only**

The last row is the project's own `CHEMBL_ASSAY_LINK_EXCLUSIONS` (`CHEMBL5133661`, `CHEMBL1074487`), reused verbatim and applied on this path too. Their aggregate counts were retained, but the original report incorrectly claimed their individual rows were exported. The follow-up now supplies all four in `followup/out/known_bad_excluded_records.csv`. ChEMBL's `data_validity_comment` and `potential_duplicate` flags are **recorded per measurement and reported, not used to drop rows**, matching how the production pipeline stores them. Here "valid exact" means eligible under these analysis rules; it is not an independent certification of experimental quality.

Why exclude organism-format observations? A whole-organism response may combine several targets, metabolism and exposure, so it cannot automatically support a direct protein-binding edge. The original gate is nevertheless too broad to justify discarding the context: 21,937 of its 21,999 excluded rows have

organism targets, but **36 have protein, family or complex targets** and 26 have other assignments. The 36 include receptor occupancy in rat brain and descriptions that conflict with the organism-format label. They are now preserved and individually reviewed in `followup/out/organism_reviewed.csv`. Section 6 distinguishes contextual retention from numerical pooling.

Why require nM? nM means nanomolar, a concentration unit. The original analysis trusts ChEMBL's standardisation and accepts only literal nM for numerical summaries. It nevertheless exports all **2,714** non-nM records with status `excluded_units`. The earlier "not kept" label was wrong. Some concentrations are convertible; dose, percentage, rate and missing units need different treatment. The follow-up finds **30** explicit concentration conversions and **2,334** conditional mass-based conversions. The latter depend on the material and molecular-weight basis, so remain a separate sensitivity scenario. No conversion changes catalogue membership.

The retained relation counts are **80,812** >, 902 <, 87 >=, 33 <=, 28 ~, and 5 >>. The legacy status name `retained_as_bound` includes the 28 approximate results; none enters an exact median. A bound can constrain potency without identifying its exact value and cannot universally be labelled "tested, no activity detected".

Claim granularity

A claim is `(molecule parent, target, role)` where role is `curated`, `binding` or `functional`. Endpoint types (Ki, Kd, binding IC50, functional IC50, EC50, AC50, Kb) are separate classes and are **never pooled**; replication is only ever counted inside one class. Two binding studies do not establish agonism. A well-evidenced target does not upgrade any other claim for the same molecule.

Tier distribution

Preliminary means a claim has a qualifying exact measurement linked to a study, but lacks the report's required replication within the same endpoint class. Usually it is one paper; it can also be a patent, a deposited dataset, or several papers split across endpoints. Twenty measurements from one paper remain one study. This is an evidence category for one molecule-target-role claim, not a clinical development phase or a verdict on the whole molecule. Species and target-assignment specificity remain separate: a Preliminary claim is not automatically a direct human-protein claim.

Role	Supported	Preliminary	Limited/unresolved
curated	1,859	-	140
binding	4,441	11,753	70,804
functional	2,714	26,753	20,045
All	**9,014**	**38,506**	**90,989**

Restricted to **direct human single-protein** evidence - the report's strict comparison for a named human protein. Human family and complex claims remain valid at their asserted level and are compared separately in the follow-up:

Role	Supported	Preliminary	Not applicable
curated	1,195	-	804
binding	1,980	4,116	80,902
functional	267	683	48,562
All	**3,442**	**4,799**	**130,268**

Supported drops from 9,014 to 3,442 - **62% of Supported claims lose that status when evidence outside this strict basis, including family and complex claims, is set aside.** Both columns are reported; neither replaces the other.

Why so much is Limited or unresolved

Reason	Claims
`censored_only` - no qualifying exact value and at least one bound/approximate result	**59,327**
`null_value_only`	30,735
`no_qualifying_exact_measurement`	494
`invalid_units_only`	293
`curated_no_target_assigned` - `drug_mechanism.tid` is null	139
`curated_untraceable` - no reference and no comment	1

The reason is preserved on every claim rather than discarded. 63,545 claims carry at least one censored measurement. A failed fetch is never recorded as a negative finding, and missing evidence never becomes an interaction edge.

The two Supported routes, and their intersection

The routes land on different claim rows by construction - curated support is a property of a curated claim, replication of an experimental one - so the intersection is a **molecule-target** fact and is rolled up as one (`out/target_pairs.csv.gz`):

Support route	Pairs
Curated only	1,321
Replicated measurement only	6,138
Both	**538**
Neither	118,199

7,997 pairs carry any support at all, covering 1,555 molecules and 2,090 targets. Only **6.7% of supported pairs are supported both ways.**

Two caveats the numbers cannot carry on their own:

- **Curated support does not imply an independently replicated measurement.** It is one curator's assertion with a traceable reference. 2,263 of the 2,267 mechanism records carry at least one; ChEMBL's

references include DailyMed, FDA labels and Wikipedia as well as PubMed.

- **Multiple papers are a documented proxy for independence, not proof of it.** Two publications may share a laboratory, a compound batch or an assay protocol. What is measured is "two distinguishable publications", and that is what the column is named.

Deposited bulk datasets never create multi-study support, and the scale of what that prevents is worth stating plainly. The 117,519 valid exact measurements resolve to **19,321 distinct studies**, of which 18,855 are publications:

Provenance	Exact measurements	Distinct studies
Publication (PubMed id or journal DOI)	89,829	18,855
Deposited dataset	26,095	**45**
Patent	1,592	-
Unresolved	3	-

26,095 measurements collapse to 45 studies, and 19,041 of them - the Sanger Genomics of Drug Sensitivity in Cancer deposit - collapse to exactly **one**. Counting activity rows, or dataset DOIs, as studies would have manufactured replication on that scale. Validation check 5 asserts zero claims reached Supported that way.

Target-assignment specificity

Assignment describes what the experiment can be attributed to. **Homologous** means a related protein is assigned in place of a confidently identified exact target, often across species. When the original species is unknown, even a human-labelled target record may carry a homologous assignment. Scores 4, 6 and 8 distinguish multiple homologous proteins, homologous complex subunits and a homologous single protein; relationship `H` also marks homology.

Non-protein or unassigned groups several different cases: confidence 0 / U means not yet assigned or curated; 1 / N means a non-molecular target such as a cell line or organism; 2 / S means a subcellular fraction; and 3 / M means a molecular target other than a protein. Such observations can be useful without identifying which named protein caused a response. ChEMBL assay definitions

Assignment	Claims
Homologous or unknown species (confidence 4/6/8, or relationship H)	78,928
Non-protein or unassigned (confidence <= 3, relationship N/S/M/U)	25,499
Human direct single protein (type SINGLE PROTEIN, tax 9606, confidence 9, relationship D, no variant)	**22,003**
Nonhuman	9,001
Human curated	1,738
Human family or multiple (confidence 5, or PROTEIN FAMILY)	571
Human direct complex (confidence 7)	430
No target assigned	139
Human direct single protein, **mutant** (variant present)	49

ChEMBL confidence 9, 7 and 8 are three *different assignment categories* - direct single protein, direct complex subunits, homologous single protein - not a general experimental-quality score. **A generic** `>= 7` **filter mixes all three, including homologous score-8 assignments.** The earlier claim that it would admit all 78,928 rows of the combined 4/6/8 category was incorrect. This report never uses `>= 7` as a proxy; it tests for 9 + D + SINGLE PROTEIN + tax 9606 + no variant explicitly, and reports the rest separately.

Family and complex claims are not fanned out

477 curated claims are asserted at family or complex level and stay there:

Molecule	Target	ChEMBL type	Action
PRAZOSIN	Adrenergic receptor alpha-1	PROTEIN FAMILY	ANTAGONIST
NICOTINE	Neuronal acetylcholine receptor α4/β2	PROTEIN COMPLEX	AGONIST
DIAZEPAM	GABA-A receptor; anion channel	PROTEIN COMPLEX GROUP	POSITIVE ALLOSTERIC MODULATOR
FLUMAZENIL	GABA-A receptor; anion channel	PROTEIN COMPLEX GROUP	ALLOSTERIC ANTAGONIST
INDOMETHACIN	Cyclooxygenase	PROTEIN FAMILY	INHIBITOR

Expanding these to their member components would have produced **3,146 claims instead of 477 - 2,669 additional "supported" claims carrying no additional assertion.** That is exactly the mechanism-fan-out failure the data-quality log records as detected by nothing in the production pipeline, quantified here. Validation check 8 asserts it did not happen.

Conflicts

- **Directional action conflicts: 0.** No molecule-target pair carries curated actions that disagree in direction (POSITIVE vs NEGATIVE modulator). Independently confirmed against ChEMBL directly, so this is a real zero, not a broken check.
- **Action refinements: 5.** Different action strings that do not contradict - varenicline `AGONIST` / `PARTIAL AGONIST` at the α4β2 nicotinic receptor, fingolimod `AGONIST` / `BINDING AGENT`, chloroquine `BINDING AGENT` / `INHIBITOR`. Counted separately so a refinement is never reported as a contradiction.
- **Measurement dispersion: 1,832 claims** where values within one endpoint class, from two or more distinct studies, span more than 100×. Prazosin at the α1 adrenergic receptor spans 0.01 nM to 1,148 nM in binding Kd across three studies - a 100,000-fold disagreement inside one supported claim.

A conflict never raises a tier. A claim with a conflict keeps its tier and carries the flag and detail, so accumulating incompatible reports cannot look like stronger support.

Threshold sensitivity

Applied **only inside the fixed candidate set**. Thresholds admit no molecule and remove none, and say nothing about physiological exposure. Curated weak relationships keep their membership and their tier at every threshold - caffeine's curated ADORA3 mechanism at Ki = 100 µM is unaffected.

Aggregation rule: a `(claim, endpoint class)` passes at T when the **log-space median** of its qualifying exact nM values is <= T. The looser "any measurement <= T" rule is reported alongside.

Threshold	All evidence: claims (median / any)	All: molecules	Human-direct: claims	Human-direct: molecules
1 µM	20,828 / 22,075	1,164 / 1,183	4,864 / 5,158	799 / 825
10 µM	33,885 / 34,671	1,296 / 1,301	6,974 / 7,118	917 / 928
100 µM	44,843 / 44,996	1,355 / 1,359	7,972 / 8,003	991 / 992
300 µM	48,321 / 48,425	1,362 / 1,365	8,161 / 8,183	1,007 / 1,010

The choice of threshold matters far less than the choice of evidence basis. Moving from 1 µM to 300 µM changes the molecule count by 17% (1,164 -> 1,362); moving from all evidence to direct human single-protein evidence changes it by 31% at 1 µM and, more importantly, cuts qualifying claims by 77%. The median and "any" rules barely differ - this data is not threshold-sensitive in the way the brainstorming document anticipated.

Target and family coverage

4,885 distinct ChEMBL target records, representing 3,750 UniProt accessions; **1,676 target records are typed as human single proteins**. A human single protein means one specified protein type, such as human ADORA1, rather than a family, multi-protein complex, cell line or organism. It does not mean one physical protein molecule or an experiment in human participants: a human receptor expressed in a laboratory cell system can qualify.

Target identity alone is weaker than **direct human single-protein evidence**, which additionally requires confidence 9, relationship D and no recorded variant. The 1,676 count does not mean that every target has direct experimental support or is relevant to the brain. Against the current graph: **206 of the present 207 targets reappear, and 3,544 accessions are new.**

ChEMBL protein class	Targets	Claims	Supported	Not in current graph
Enzyme	1,737	42,956	2,586	1,662
(unclassified)	1,495	26,749	2,568	1,495
Membrane receptor	524	47,974	2,051	445
Ion channel	306	7,434	692	237
Transporter	159	4,850	372	146
Transcription factor	125	5,877	251	119
Epigenetic regulator	86	355	119	84

Family labels come from ChEMBL's `protein_classification`, used **only** to summarise coverage. The data-quality log records that it must never be used to infer physical assembly - it is a sequence and function taxonomy blind to that, filing CACNA1C and CACNA1D on one leaf.

Best-evidenced human single-protein targets:

Target	ChEMBL	Molecules	Supported claims
D(2) dopamine receptor	CHEMBL217	745	99
5-HT2A receptor	CHEMBL224	735	83
hERG (KCNH2)	CHEMBL240	859	75
μ-opioid receptor	CHEMBL233	858	59
Noradrenaline transporter	CHEMBL222	847	59
Serotonin transporter	CHEMBL228	846	58
5-HT1A receptor	CHEMBL214	826	53

This is a genuinely different centre of gravity from the present graph, whose best-connected nodes are the adenosine receptors, the phosphodiesterases and the cholinesterases. hERG's position is a caution, not a discovery: it is a safety-screening target, and its 859 molecules reflect regulatory panels, not brain pharmacology. **Retaining peripheral off-targets with their context - rather than labelling every target of a clinically selected drug a CNS mechanism - is why hERG appears here honestly instead of being quietly promoted.**

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4. The existing 28

No grandfathering. Each seed was tested against the same rules.

25 of 28 qualify. 15 by both routes, 10 by indication only, 3 not admitted. 24 of the 25 have at least one Supported direct-human claim.

Seed	In?	ATC N	CNS ind.	Admission / exclusion reason
caffeine	yes	yes	yes	ATC N (N06BC01); 18 core MeSH terms
theophylline	yes	-	yes	core CNS indication (2 terms); parent `CHEMBL190`
theobromine	no	-	-	**no ATC N, and no indication record inside the considered CNS scope**
istradefylline	yes	yes	yes	ATC N (N04CX01); 3 terms
rolipram	yes	-	yes	core CNS indication (5 terms)
cilostazol	yes	-	yes	core CNS indication (7 terms)
donepezil	yes	yes	yes	ATC N (N06DA02); 24 terms
pentoxifylline	yes	-	yes	core CNS indication (8 terms)
sildenafil	yes	-	yes	core CNS indication (13 terms)
verapamil	yes	-	yes	core CNS indication (3 terms)
rivastigmine	yes	yes	yes	ATC N (N06DA03); 17 terms
tacrine	yes	yes	yes	ATC N (N06DA01); 2 terms
selegiline	yes	yes	yes	ATC N (N04BD01); 10 terms
cannabidiol	yes	yes	yes	ATC N (N03AX24); 43 terms
haloperidol	yes	yes	yes	ATC N (N05AD01); 16 terms
clozapine	yes	yes	yes	ATC N (N05AH02); 7 terms
amitriptyline	yes	yes	yes	ATC N (N06AA09); 6 terms
loperamide	no	-	-	**indications fall only in the `general_pain` class (D010146)**
dextromethorphan	yes	yes	yes	ATC N (N07XX59); 19 terms
memantine	yes	yes	yes	ATC N (N06DX01); 32 terms
lamotrigine	yes	yes	yes	ATC N (N03AX09); 21 terms
dipyridamole	yes	-	yes	core CNS indication (2 terms)
vardenafil	yes	-	yes	core CNS indication (2 terms)
dronabinol	yes	-	yes	core CNS indication (20 terms)
phenelzine	yes	yes	yes	ATC N (N06AF03); 1 term
quinidine	yes	-	yes	core CNS indication (6 terms)
galantamine	yes	yes	yes	ATC N (N06DA04); 16 terms
mibefradil	no	-	-	**indications fall only in the `cns_oncology` class (D016543)**

Seed	Claims	Sup.	Prelim.	Cur.	Multi	HD sup.
caffeine	207	21	34	1	20	10
theophylline	212	27	22	3	24	6
theobromine	-	-	-	-	-	-
istradefylline	50	8	5	1	7	6
rolipram	58	17	12	1	16	5
cilostazol	163	5	13	1	4	2
donepezil	141	21	31	1	20	8
pentoxifylline	157	2	10	2	0	0
sildenafil	202	23	30	1	22	8
verapamil	244	33	61	1	32	5
rivastigmine	24	11	6	1	10	2
tacrine	192	19	42	1	18	2
selegiline	170	9	7	1	8	3
cannabidiol	123	16	73	1	15	6
haloperidol	269	50	79	2	48	21
clozapine	256	53	80	2	51	24
amitriptyline	195	8	71	2	6	7
loperamide	-	-	-	-	-	-
dextromethorphan	170	5	25	2	3	3
memantine	104	11	15	1	10	1
lamotrigine	174	6	18	1	5	2
dipyridamole	169	8	33	2	6	2
vardenafil	51	9	19	1	8	5
dronabinol	30	12	16	1	11	6
phenelzine	140	3	22	1	2	2
quinidine	108	8	43	1	7	2
galantamine	150	10	7	1	9	3
mibefradil	-	-	-	-	-	-

Each exclusion fails for a *different, documented* reason, and each is a fair test of a boundary decision:

- **theobromine** has neither an ATC N code nor any CNS indication in ChEMBL - a genuine miss. It is a methylxanthine with real ADORA pharmacology, and the clinical rule cannot see it. This is the clearest evidence that a clinical-only policy is insufficient on its own.

- **loperamide** is excluded only because general pain is outside the core scope. Under variant S4 it would be admitted. It has a curated μ-opioid agonist mechanism, and it is the only route in the current graph to OPRD1/OPRK1/OPRM1 - an expensive exclusion made by a boundary decision, not by evidence.
- **mibefradil** is excluded only by the CNS-oncology decision; under S1 it would be admitted. It is also the only molecule in the graph measuring CACNA1G/1H/1I individually.

Two of three exclusions are reversible by a single documented boundary choice. That is worth weighing against the fact that both boundaries are defensible on their own terms.

Note also **pentoxifylline**: in the catalogue, 157 claims, but **0** multi-study support and **0** supported direct-human claims - its two curated mechanisms are family-level PDE assertions. It is in the graph today drawing 13 solid edges from one curator's family-level claim.

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5. Coverage gaps and limits

Molecules retained with no evidence

361 catalogue molecules have no assessable ChEMBL 37 evidence at any target. They stay in the catalogue as coverage gaps; no edge was manufactured for them.

By type: 204 small molecules, 65 Unknown, 28 Protein, 19 oligonucleotide, 17 antibody, 16 gene, plus 12 others. By route: 272 indication-only, 47 ATC-only, 42 both.

Seven carry an Approved CNS indication but have no assessable target evidence under these ChEMBL 37 report rules: `MEPHOBARBITAL`, `MEDIFOXAMINE`, `FONAZINE`, `THIOPROPAZATE`, `PHENETURIDE`, `ACEGLUMATE`, and `GINKGO` (a botanical with no structure). Older drugs and botanicals are systematically under-represented in a modern screening database - the absence is about ChEMBL, not about the pharmacology.

A further **116 molecules have claims but none above Limited or unresolved.**

What this report did not assess

Not assessed	Consequence
PDSP, BindingDB, DGIdb	The project's supplementary sources. Their richer coverage of the existing 28 is **not** extrapolated to the other 2,217 - doing so would let familiarity masquerade as biology.
GtoPdb, HPA, Chemical Probes Portal, B3DB	Outside the original report and this follow-up. Check current access terms and source coverage before any integration.
Brain expression, BBB penetration, exposure	Kept as context, never a gate. Absence of annotation remains unknown, per invariant 10.
Literature beyond ChEMBL	Out of scope entirely.

Every "no evidence" statement means no eligible evidence under the stated rules and source coverage. It does not mean no published pharmacology, and a gap under the direct-human comparison may still have other ChEMBL evidence. The recorded molecule/source combinations are exactly: all 2,245 candidates × ChEMBL 37, assessed; all 2,245 × every other source, **not assessed**.

Measured versus estimated

Baseline counts come from ChEMBL 37 and the original MeSH snapshot, with supporting rows in `analysis/cohort1/out/` and aggregate tables in `out/report_tables.json`. Follow-up counts and policy comparisons come from `followup/out/`. Proposed mass conversions are conditional calculations using an explicitly stated material assumption; they are not newly measured values. Where a source or additional cohort was not assessed it remains unknown or not assessed, never zero. The graph has not been benchmarked at the proposed catalogue sizes.

Unresolved mappings

- MeSH: **0 of 2,178** ChEMBL descriptor ids failed to resolve.
- Identity: 222 candidates have no structure (`structure_type = NONE`) and are marked unresolved rather than guessed at.
- Targets: 139 curated mechanism records have no target assigned; they are kept as Limited/unresolved with that reason, not dropped.
- 1,495 observed targets carry no ChEMBL protein-class assignment and are reported as `(unclassified)` rather than being forced into a family.

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6. Follow-up findings and proposed import policy

No policy is adopted or implemented by this report. The September 12 follow-up answers the Zotero questions and measures concrete choices using the original candidate and evidence exports. Full methods, conversion audit, all 28 seeds under every policy, and limitations are in COHORT-FOLLOWUP.md and `analysis/cohort1/followup/`.

1. Count both admission routes explicitly

The earlier recommendation quoted **997** while also recommending retention of both routes. That is the indication-only Approved/Phase-3 count. Keeping whole ATC N alongside it gives **1,206** candidates, or **995 small molecules**. An ATC-only candidate remains eligible even when its CNS phase is missing; this does not label it CNS-approved.

Selection	All types	Small molecules	Seeds in small-molecule set
Whole clinical catalogue	2,245	1,771	24
CNS Approved/Phase 3 only	997	791	22
ATC N + CNS Approved	761	677	18
ATC N + CNS Approved/Phase 3	1,206	995	22
ATC N + CNS Approved/Phase 2/3	1,912	1,540	24

These use S0 core indications. All 28 seeds are tested without grandfathering. The small-molecule definition follows the parent's source type literally. Both theophylline records, including parent `CHEMBL190`, are typed `Protein` in ChEMBL 37. That source anomaly costs theophylline under a literal small-molecule filter even before considering its Phase-1 CNS indication. It needs an explicit identity correction, not a

silent exception.

2. Separate catalogue membership from drawable evidence

The following uses the **original baseline evidence**, keeping supported human single proteins, precise complexes, families and complex groups at their native ChEMBL target level. Experimental support is recalculated only from direct human assignments within each accepted type; mutants and homologous assignments do not contribute. Curated support retains its own route and is not experimental replication.

"Usable" means at least one Supported or Preliminary claim. A pair is a molecule-target pair, which may have several role-specific claims. Targets are ChEMBL target records, not counts of expanded component proteins.

Small-molecule selection	Supported molecules	Supported pairs	Targets with support	Usable molecules	No qualifying edge
Whole clinical catalogue	1,137	3,074	795	1,310	461
CNS Approved/Phase 3 only	607	1,528	426	665	126
ATC N + CNS Approved	448	1,145	264	481	196
ATC N + CNS Approved/Phase 3	666	1,653	431	738	257
ATC N + CNS Approved/Phase 2/3	1,013	2,738	749	1,152	388

For the same 995 candidates, target granularity changes what can be shown:

Evidence basis for the same 995 candidates	Supported claims	Supported molecules	Usable molecules
Human single proteins	1,719	532	641
+ precise protein complexes	1,804	557	663
+ families and complex groups	2,049	666	738

My proposed starting catalogue is **ATC N plus CNS Approved/Phase 3, small molecules first**. Keep all 995 catalogue entries. Its broad human-specific evidence view reaches **666 molecules with Supported evidence** and **738 with Supported or Preliminary evidence**; the other **257 remain visible as coverage gaps** for this basis. Those are not necessarily gaps in all ChEMBL evidence.

Families and complexes need their own target nodes or rows. Their evidence must not be distributed to subunits or counted as several independent shared targets. Show Supported claims initially, offer Preliminary and other-species evidence as explicit layers, and keep curated versus multi-study support visible. Clinical membership does not certify a CNS mechanism for every edge.

This candidate set carries **63 family withdrawal flags** and **12 records without a structure**. Preserve those annotations; "clinical catalogue" does not mean currently available or uniformly resolved.

3. Treat the Phase-2 expansion as a real tradeoff

Moving from Approved/Phase 3 to Approved/Phase 2/3 while keeping ATC N adds **545 small molecules**, **347 molecules with Supported evidence**, and **318 targets with support** under the broad human-specific basis above. It also retains rolipram and pentoxifylline, which the Phase-3 choice loses. The original claim that this necessarily adds thinner evidence was not established by its stage counts and is withdrawn. The choice is between a tighter clinical-stage scope and substantially wider target coverage.

4. Preserve excluded evidence before broadening numerical rules

The unit audit compares explicit concentration conversions with a conditional scenario that also uses the measured compound record's full molecular weight. The scenarios are alternatives; their counts must not be added together.

Recovery scenario	Assignment basis	Added measurements	New Preliminary claims	New Supported claims
Concentration conversions	All assignments	30	18	1
Concentration conversions	Direct human single protein	6	1	1
Including conditional mass conversions	All assignments	2,364	243	106
Including conditional mass conversions	Direct human single protein	35	7	1

The direct-human upgrade from explicit concentration conversion is **phenelzine-MAO-A binding**; imipramine-BRD4 binding gains a Preliminary claim. The latter's converted values are 647 and 1,020 micromolar, demonstrating why an exact value and evidence tier do not imply useful potency or exposure. Conversion does not independently validate upstream units or target species.

The 2,334 mass-concentration proposals remain conditional until the original material basis is checked. Another 350 non-nM rows remain unrecovered. No existing coverage-gap molecule gains its first assessed claim through either conversion scenario. In the proposed 995-molecule catalogue, explicit concentration recovery adds one Supported claim without adding another molecule with qualifying evidence.

All **21,999 organism-format exclusions** are now available in a context inventory. All **36 protein-assigned records**, spanning 22 assays, have been reviewed from their descriptions. **31** have positive exact nM values, but their contexts include occupancy, downstream physiology, cell assays with conflicting format metadata, and microorganism screens. Keeping this context causes **zero changes to the baseline biochemical tiers**. Blindly pooling the 31 values would produce 18 new Preliminary and 2 new Supported claims under the old counting rule; that is a diagnostic of the risk, not a recommended evidence upgrade.

5. Keep scope extensions separate from evidence claims

Core plus CNS oncology plus general pain gives **2,523 molecules (+278)**, not the earlier 2,445 figure, which counted core plus pain alone. The widest variant is 2,611 (+366). Neither delta is a count of useful new graph edges: evidence for molecules added outside the original 2,245 was not assessed. The corrected scope counts alone do not justify adopting either extension.

A clinical-only catalogue still misses theobromine and other nonclinical ligands. A future curated-ligand route remains a separate candidate analysis, with explicit eligibility and identity rules; its coverage is not

measured or assumed here. Loperamide and mibefradil should be judged by those explicit routes or a documented scope choice, not simply because they are old seeds.

6. Build the viewer around selected neighbourhoods

The table measures candidates, claims, pairs and targets; it does not benchmark rendering performance. A graph need not draw every catalogue item at once. Searchable membership, target-level identity, selectable neighbourhoods and clear evidence filters should precede a production import. The original 2,245-by-4,885 product was not an observed edge count and did not establish that a particular graph implementation was impossible to render.

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7. Validation

The ten original baseline checks pass (`analysis/cohort1/out/validation.csv`). The follow-up adds eight data validations, including reproduction of baseline tiers and unchanged production fingerprints. The combined unit suite passes **29 tests**: 16 original and 13 follow-up tests. These checks validate the implemented rules and accounting; they do not certify every upstream assay assignment or the biological interpretation of a threshold.

#	Check	Result
1	Baseline reproduction - 521 / 361 / 15	**PASS**, exactly
2	Parent/salt deduplication - 2,245 distinct parents, 1,423 forms folded in, `CHEMBL1355736` -> `CHEMBL190` as one row	**PASS**
3	Admission-route union arithmetic across 13 stages	**PASS**, 0 violations
4	Unrelated-indication approval - 1,147 family-Approved vs 506 CNS-Approved, 591 divergent, never merged	**PASS**
5	Missing study provenance - 0 claims reached Supported without >= 2 publication-class studies	**PASS**
6	Multiple measurements from one paper - `CHEMBL14` vs `CHEMBL320`, 20 Ki measurements across 20 assays in one document, recorded as **1** study	**PASS**
7	Cross-source duplicates - 10 PubMed ids carried by more than one ChEMBL doc record collapse to one study; 74,691 of 106,380 links are PubMed-keyed	**PASS**
8	One family claim vs many subtype claims - 477 stay 477, not 3,146	**PASS**
9	Report totals match the exported records - 7 cross-totals	**PASS**, 0 mismatches
10	Production artifacts unchanged	**PASS**

On check 10: `git status` reports no modification to `db/schema.sql`, `scripts/build_db.py`, `scripts/chembl_local.py`, `scripts/export_graph.py` or `viz/index.html`. `db/subnetwork.db` (1,437,696 bytes) and `viz/graph.data.js` (2,330,003 bytes) both retain their pre-existing mtimes; no rebuild was run and both were opened read-only, as was the 30.5 GB ChEMBL release.

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8. Files

Schemas, join keys and per-file row counts: `analysis/cohort1/README.md`.

Rules (versioned)	`analysis/cohort1/config/cns_indication_scope.json`, `config/tier_rules.json`
Scripts	`analysis/cohort1/scripts/step1…step5`, `run_all.sh`
Source manifest	`analysis/cohort1/MANIFEST.json` - URLs, retrieval dates, checksums, access status, licences
Catalogue	`out/candidates.csv`, `candidate_atc_codes.csv`, `candidate_indications.csv`, `family_members.csv`, `indication_scope.csv`, `route_arithmetic.csv`, `stage_breakdown.csv`
Evidence	`out/assessed_claims.csv.gz`, `target_pairs.csv.gz`, `claim_studies.csv.gz`, `claim_thresholds.csv.gz`, `measurements.csv.gz`, `molecule_evidence.csv`, `observed_targets.csv`, `family_coverage.csv`, `tier_distribution.csv`, `threshold_sensitivity.csv`
Gaps and exclusions	`out/coverage_gaps.csv`, `evidence_exclusions.csv`
The 28	`out/seed_membership.csv`, `seed_report.csv`
Baseline aggregates	`out/report_tables.json`
Follow-up	`followup/out/summary.json`, `followup/out/policy_comparison.csv`, `followup/out/unit_recovery_summary.csv`, `followup/out/organism_reviewed.csv`
Validation	`out/validation.csv`, `followup/out/validation.csv`, `tests/test_cohort_report.py`, `tests/test_cohort_followup.py`

Sources. ChEMBL 37 bulk release (`/mnt/data/chembl/chembl_37/chembl_37_sqlite/chembl_37.db`, 30,480,314,368 bytes, CC BY-SA 3.0, read-only) and MeSH tree numbers from the NLM MeSH SPARQL endpoint (current-year graph, retrieved 2026-09-07, public domain, original `cache/mesh_tree.tsv` sha256 `dc8f2681…`). The original run recorded no fetch failures. Those raw MeSH cache files are now absent; their exported descriptor classification remains available and is the follow-up's scope input. Follow-up input hashes, rules and validations are recorded separately in `analysis/cohort1/followup/out/summary.json`.